

Individual Portfolio and Contribution Record

STUDENT	Sonam Pelden
STUDENT ID	CIHE23321
GROUP	5
PROJECT TITLE	Healthcare Integration Tes
ROLE IN GROUP	Architecture / design lead
SECTIONS OWNED	Yet to discuss
GITHUB REPOSITORY	https://github.com/SkinnySeal17/HEALTHCARE_SYSTEM.git
PROJECT BOARD	https://student-team-siuckpgv.atlassian.net/jira/software/projects/HITH/boards/34/backlog?epics=visible&atlOrigin=eyJpIjoiOWU2MGVIZjkyNzI2NDg2YmEzODU0ZDgxOTY3Y2E3NDUiLCJwIjoiY9
GENERATED	13 Sept, 19:50

ACTIVITY SUMMARY — GENERATED FROM THE LOG, NOT SELF-REPORTED

MEASURE	WEEK 6	WEEK 7	WEEK 8	TOTAL
Entries logged	5/5	6/6	6/7	18/20
Minutes on task	55	51	34	144
Evidence items	5	6	8	19
Completeness	100%	100%	90%	96%

Time recorded during scheduled class: 83 minutes. Time recorded outside class: 62 minutes.

1 · Individual responsibility summary

ASSESSMENT SECTION	INDIVIDUAL RESPONSIBILITY	EXPECTED OUTPUT	TARGET
Section 1 and Section 2	Business Context, Problem Statement, Goals/Objectives, Scope, Stakeholders Project Charter, Roadmap/Gantt and RACI	completed draft of the report's early sections — the Business Context, Problem Statement, Goals/Objectives, Scope, and Stakeholders section, plus the Project Charter, Roadmap/Gantt Chart, and RACI Matrix — ready for team review and integration into the final Assessment 2 report.	week 6
Section 3	Current State Assessment: As-Is Analysis Technology and Market Survey	Expected to deliver the Current State Assessment (As-Is Analysis) describing MedTalk's existing integration workflow and limitations, and the Technology and Market Survey reviewing relevant FHIR/terminology validation tools and standards — both feeding into Section 3 of the report.	week 6
Section 4	Proposed Solution Architecture Technology Stack and Justification	Expected to deliver the Proposed Solution Architecture (diagram and explanation of system components, data flow, and integrations) and the Technology Stack and Justification, forming the technical core of Section 4.	week 6

ASSESSMENT SECTION	INDIVIDUAL RESPONSIBILITY	EXPECTED OUTPUT	TARGET
Section 5	Sprint Plan Proof of Concept Description	a defined implementation approach week 6 (sprint activities, deliverables, tools) plus an initial description of how the proof-of-concept will demonstrate the test harness's technical feasibility – both feeding into Section 5 of the report.	

Week 6

5 of 5 entries logged · 55 minutes recorded · 5 evidence items · 100% complete

W6-1 · Frame the problem in your own words

Logged late · 5 min · 3/3 required parts · 13 Sept, 19:50

The problem, as you understand it

FAZ Australia's MedTalk platform relies on manual checks to validate its HL7 FHIR integrations and clinical terminology mappings against standards such as SNOMED CT and ICD-10. As MedTalk adds new hospital and clinic integrations, this manual process becomes increasingly slow, inconsistent, and vulnerable to human error, since developers must individually cross-check sample records against terminology sets and integration specifications.

The consequences are significant in a healthcare context. An incorrect terminology mapping, such as a wrongly linked SNOMED CT code, can distort a patient's clinical record and compromise data accuracy and patient safety. It also exposes FAZ Australia to compliance risk under Australian healthcare data standards, since there is currently no systematic way to verify that integrations meet required conformance rules before deployment.

This project addresses that gap by designing an automated Healthcare Integration and Clinical Terminology Test Harness that validates FHIR-based message conformance and terminology mappings prior to release, reducing manual effort and the risk of undetected data errors reaching clinical systems.

One thing that is deliberately OUT of scope, and why

Real-time production monitoring of live patient data is deliberately excluded from this project's scope. While the test harness validates FHIR message conformance and terminology mappings before deployment, it does not monitor or intercept live clinical data flowing through MedTalk's production systems. This exclusion reflects both time constraints within the semester and ethical considerations around accessing real patient records without appropriate clinical governance approval. Instead, the project uses synthetic or de-identified test data to validate integration logic. Production monitoring would require ongoing operational access, hospital-grade compliance sign-off, and ethics approval beyond what is feasible for a capstone project of this duration.

Assessment 2 section you worked on today

Section 1 — Business Context, Problem Statement, Goals/Objectives, Scope, and Stakeholders

W6-2 · Literature search log

Logged · 20 min · 6/6 required parts · 31 Aug, 09:33

Search strings you actually typed

Walonoski FHIR validation testing 2018
SNOMED CT
ICD-10 SNOMED CT mapping trial

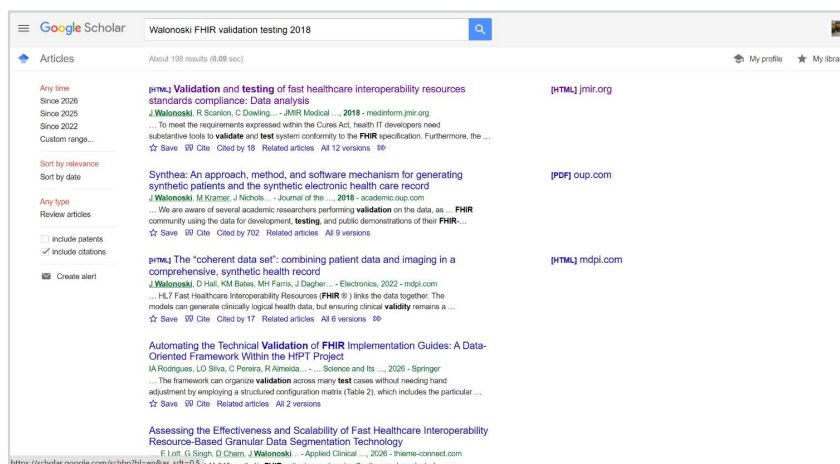
Databases / sites searched

Google Scholar

SOURCE (AUTHOR, YEAR, TITL	LINK OR DOI	KEY INFORMATION YOU OBTAIN	HOW IT WILL BE USED
JMIR Walonoski et al. (2018), Validation and Testing of Fast Healthcare Interoperability Resources Standards Compliance: Data Analysis	https://medinform.jmir.org/2018/4/e10870/	Found that regular use of FHIR validation/testing tools (Crucible, Touchstone) is strongly correlated with improved server compliance with the HL7 FHIR spec	Justifies the need for an automated test harness — supports Section 3 (Technology and Market Survey) and the project rationale
Ohlsen, Hofer & Ingenerf (2025), A Validation Tool (VaPCE) for Postcoordinated SNOMED CT Expressions	https://medinform.jmir.org/2025/1/e67984/	Describes a tool using FHIR's \$validate-code operation with a terminology server to detect and explain SNOMED CT syntactic/semantic errors	Direct technical precedent for the test harness's terminology validation module (Section 4 – Solution Design)

SOURCE (AUTHOR, YEAR, TITL	LINK OR DOI	KEY INFORMATION YOU OBTAIN	HOW IT WILL BE USED
Vikström et al. (2007), Mapping the categories of the Swedish primary health care version of ICD-10 to SNOMED CT concepts	https://link.springer.com/article/10.1186/1472-6947-7-9	Found intercoder reliability of 0.88% when manually mapping ICD-10 to SNOMED CT, showing manual mapping is error-prone	Evidence for the problem statement – quantifies risk of manual terminology mapping

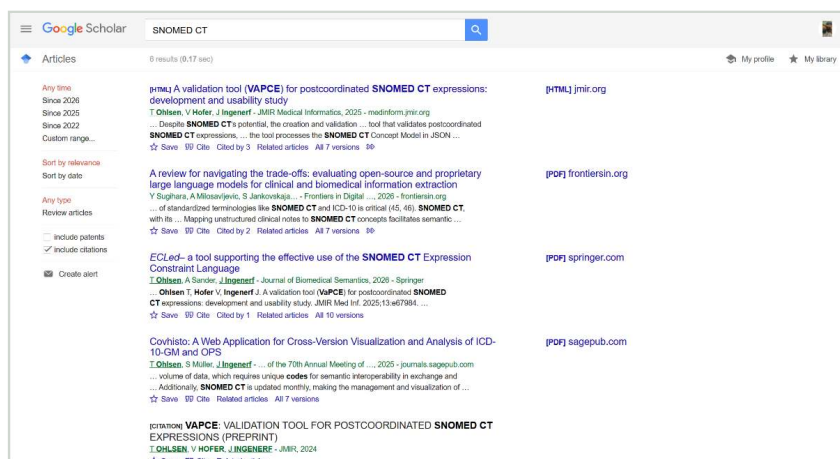
W6-2-01 · 24 AUG 2026



Section: Section 3 Technology and Market Survey) & Introduction/rationale · Task: Literature search log

What it demonstrates: the screenshot shows the file that i have searched

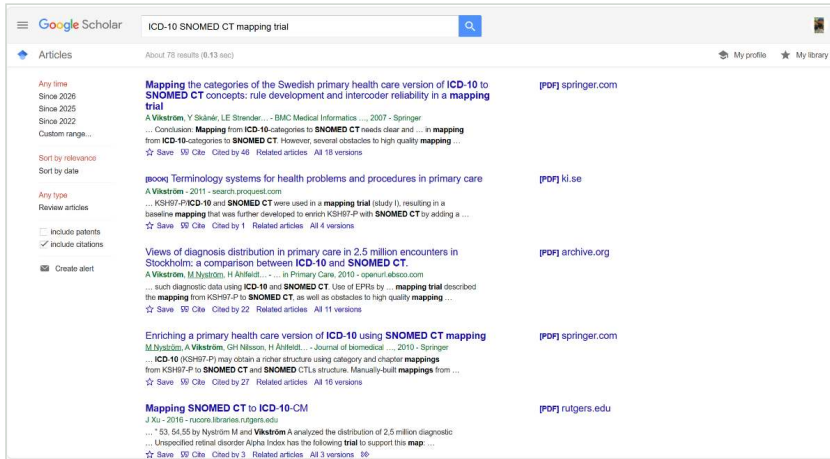
W6-2-02 · 31 AUG 2026



Section: Section 4.1 Proposed solution Architecture · Task: Literature search log

What it demonstrates: this image shows the file related to the solution design

W6-2-03 · 31 AUG 2026



Section: section 1.1 Business Context and Problem Statement · Task: Literature search log

What it demonstrates: This image support the business context

W6-3 · Claim your tasks on the board

Logged · 4 min · 4/4 required parts · 31 Aug, 09:34

Tasks you created or took ownership of (one per line, with task ID)

Task created:

- HITH-31 – Business Context, Problem Statement, Goals/Objectives, Scope, Stakeholders
- HITH-32 – Project Charter, Roadmap/Gantt and RACI
- HITH-33 – Current State Assessment: As-Is Analysis
- HITH-34 – Technology and Market Survey
- HITH-35 – Proposed Solution Architecture
- HITH-36 – Technology Stack and Justification
- HITH-37 – Sprint Plan

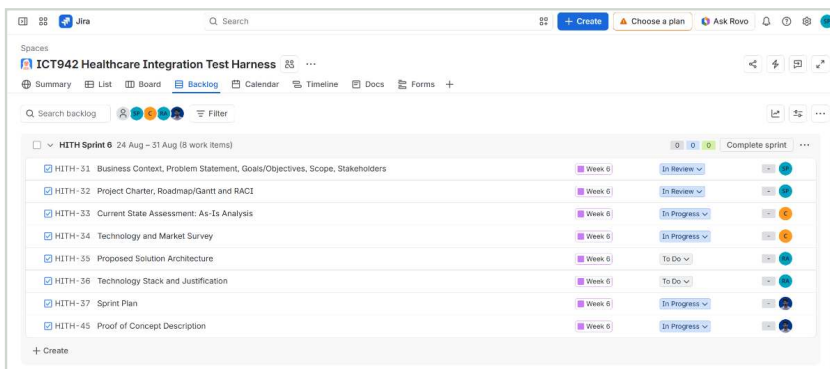
Task Assigned(ownership):

- HITH-31 – Business Context, Problem Statement, Goals/Objectives, Scope, Stakeholders
- HITH-32 – Project Charter, Roadmap/Gantt and RACI

Your contribution to the charter, scope, requirements, RACI, risk register or feasibility study

I was responsible for drafting the Business Context, Problem Statement, Goals/Objectives, Scope, and Stakeholders section, defining the cybersecurity problem for FAZ Australia's MedTalk platform and setting SMART objectives and success criteria. I also developed the Project Charter, Project Roadmap/Gantt Chart, and RACI Matrix, defining project scope, deliverables, and role responsibilities across the team. In addition, I contributed to the Risk Register and Feasibility Study, identifying key project risks and assessing technical, operational, and legal feasibility.

W6-3-01 · 31 AUG 2026



Section: All the section · Task: Claim your tasks on the board

W6-4 · Draw the first figure

Logged late · 9 min · 5/5 required parts · 13 Sept, 19:44

What you drew

Problem context / stakeholder map

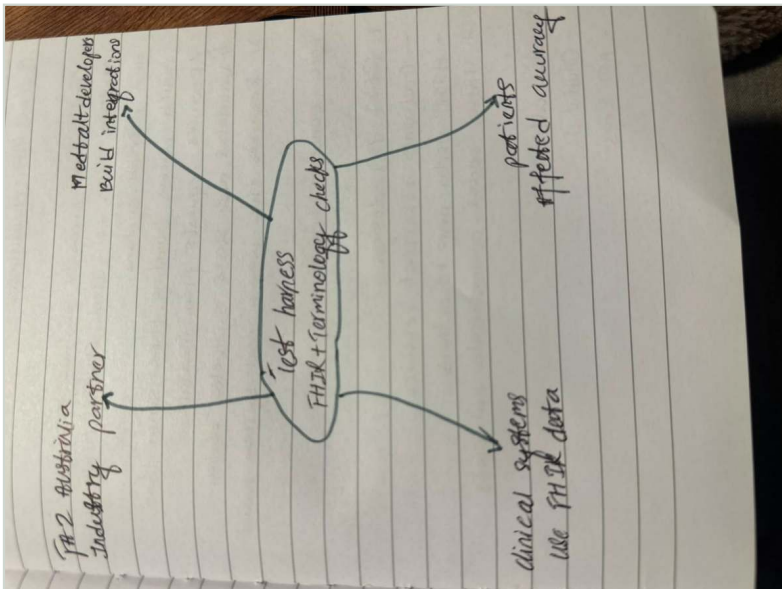
Explain the figure — what a reader should take from it

I drew a stakeholder map centred on the proposed Test Harness, showing its relationships to key stakeholders: FAZ Australia (industry partner and platform owner), the MedTalk development team (who will use the harness before releasing integrations), receiving hospital/clinic systems (who consume validated FHIR data), and patients (indirectly affected through data accuracy and safety). Lines between boxes indicate the flow of data, requirements, and validation feedback. This figure grounds the problem in its real organisational context rather than treating it as a purely technical issue, showing why validation errors matter beyond just failed tests.

One part of the figure you are not confident about

I'm not fully confident I've captured every indirect stakeholder correctly — for example, whether regulatory bodies overseeing Australian healthcare data standards should be shown as a direct stakeholder or only an influencing constraint on the project.

W6-4-01 · 13 SEPT 2026



Section: — · Task: Draw the first figure

What it demonstrates: diagram

W6-5 · Weekly reflection

Logged · 18 min · 4/4 required parts · 31 Aug, 09:50

What did I personally complete this week?

For Week 6, I have completed two key tasks: drafting the Business Context, Problem Statement, Goals/Objectives, Scope, and Stakeholders section, and developing the Project Charter, Roadmap/Gantt Chart, and RACI Matrix. These tasks required defining the cybersecurity problem for FAZ Australia's MedTalk platform, setting SMART objectives and success criteria, outlining project scope and stakeholder requirements, and formalising team roles and responsibilities to guide the project through its remaining phases.

What did I learn about the problem or the technology?

I learned that manual validation of HL7 FHIR integrations and SNOMED CT terminology mappings is a well-documented weak point in healthcare systems, not just an issue specific to MedTalk. Literature confirmed that manual terminology mapping has measurable error rates (e.g. intercoder agreement around 83% in one study), and that automated FHIR

validation tools have been shown to meaningfully improve compliance and reduce errors compared to manual checks. This reinforced that our proposed test harness addresses a real, recognised gap rather than a hypothetical one, and pointed us toward using established validation approaches (e.g. FHIR's \$validate-code operation) rather than building terminology-checking logic from scratch.

What decision did I influence, and why?

As Project Lead, I influenced the decision to scope the project around FHIR conformance and terminology validation only, deliberately excluding real-time production monitoring of live patient data. This was based on time constraints and the ethical risk of accessing real patient records without clinical governance approval. I also influenced the RACI matrix structure, assigning clear ownership of feasibility, requirements, and architecture tasks to team members based on their strengths, which helped keep Week 6 deliverables on track despite parallel workstreams.

What will I finish before the next class?

Before the next class, I completed the drafts of both my assigned Week 6 tasks: the Business Context, Problem Statement, Goals/Objectives, Scope, and Stakeholders section. Tasks are now marked as In Review on Jira, ready for team feedback and integration into the full Assessment 2 report before the next submission checkpoint.

Week 7

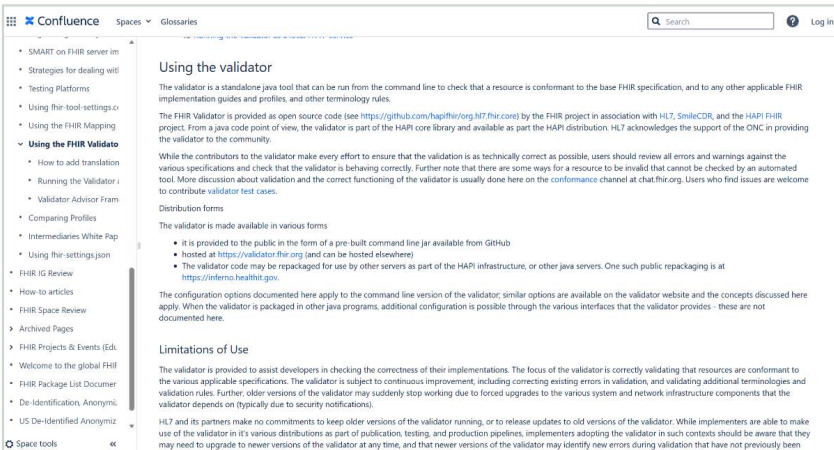
6 of 6 entries logged · 51 minutes recorded · 6 evidence items · 100% complete

W7-1 · Compare three existing tools

Logged late · 18 min · 4/4 required parts · 07 Sept, 09:41

TOOL / SYSTEM / APPROACH	VENDOR & LICENCE	FEATURES RELEVANT TO OUR P	LIMITATION WE CARE ABOUT	SOURCE
HAPI FHIR Validator	Smile CDR / HL7 community — open source, Apache 2.0 Licence	Validates FHIR resources against StructureDefinitions and profiles; uses terminology services to check codes (e.g. SNOMED CT, LOINC) against ValueSets; runs locally or via CLI, so it can be embedded directly into a CI/testing pipeline	In-memory terminology checking cannot validate external CodeSystems (like full SNOMED CT or LOINC) without connecting to an external terminology server, adding setup complexity	HL7 FHIR Confluence, "Using the FHIR Validator," https://confluence.hl7.org/display/FHIR/Using+the+FHIR+V+alidator
AEGIS Touchstone	AEGIS.net, Inc. — commercial, cloud-based subscription platform (free tier available)	Automated, cloud-based conformance and interoperability testing against the FHIR specification; supports custom TestScripts and continuous conformance measurement, useful as a benchmark for our test harness design	Commercial licensing cost and cloud dependency make it unsuitable as a directly embedded solution for a capstone-scale, self-hosted test harness	AEGIS.net, "Touchstone – Anchor of the AEGIS testing ecosystem," https://touchstone.com/
Ontoserver (CSIRO)	CSIRO (Australian e-Health Research Centre) — commercial licence, with free academic-use tier	Purpose-built FHIR terminology server with deep SNOMED CT-AU support; provides \$validate-code, \$expand, and \$subsumes operations for verifying terminology mappings; underpins Australia's National Clinical Terminology Service	Requires a formal licensing agreement and infrastructure to host, which adds cost and setup overhead not ideal for a lightweight capstone proof-of-concept	Ontoserver, "The gold-standard of FHIR terminology servers," https://www.ontoserver.csiro.au/site/our-solutions/ontoserver/

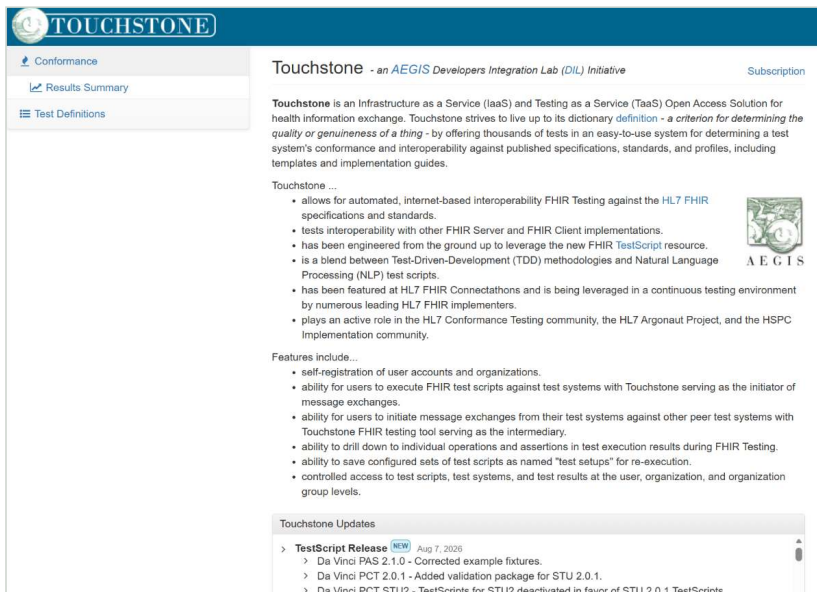
W7-1-01 · 31 AUG 2026



Section: section 3 · Task: Compare three existing tools

What it demonstrates: Shows the validator is open source (Apache 2.0), distributed as a standalone Java tool, and documents its known limitations

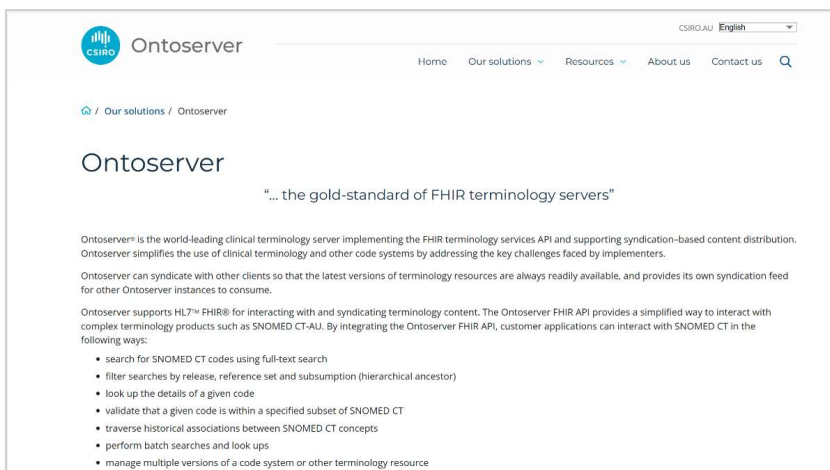
W7-1-02 · 31 AUG 2026



Section: section 3 · Task: Compare three existing tools

What it demonstrates: Demonstrates it is a cloud-based commercial FHIR conformance testing platform with automated TestScript execution

W7-1-03 · 31 AUG 2026



Section: section 3 · Task: Compare three existing tools

What it demonstrates: Demonstrates it is a commercial, SNOMED CT-focused FHIR terminology server with validation and search features

W7-2 · State the gap and your contribution

Logged late · 16 min · 4/4 required parts · 31 Aug, 11:16

The gap in existing solutions

Existing tools each address only part of the problem. HAPI FHIR Validator checks structural conformance to FHIR profiles but cannot fully verify external terminology content, like SNOMED CT, without connecting to a separate terminology server. Ontoserver handles terminology validation well but does not test FHIR message-level conformance. AEGIS Touchstone offers broad, automated conformance testing but is a commercial, cloud-hosted platform, adding licensing cost and infrastructure dependency unsuitable for a lightweight, self-hosted solution.

This leaves a clear gap: no existing tool combines FHIR conformance checking and clinical terminology validation into a single, integrated solution tailored to a specific platform's needs. FAZ Australia currently relies on manual cross-checking to bridge this gap, which is slow and error-prone. The proposed test harness addresses this by combining both validation types into one lightweight, purpose-built tool integrated directly into MedTalk's development and deployment workflow.

Your proposed technical contribution to closing it

My proposed technical contribution is to design and build a lightweight, integrated test harness that combines FHIR conformance validation and clinical terminology mapping checks into a single automated tool, tailored specifically to MedTalk's integration workflow. Rather than relying on separate tools like HAPI FHIR Validator and Ontoserver run independently, or a costly commercial platform like Touchstone, the harness will use the FHIR \$validate-code operation alongside profile-based structural validation to check both message conformance and terminology accuracy in one pass. This reduces manual cross-checking effort, catches errors earlier in the development cycle, and gives FAZ Australia a purpose-built, self-hosted validation step that fits directly into their existing release process.

Technology or framework you selected

HAPI FHIR, FHIR \$validate-code operation

One alternative you rejected, and why

I rejected Touchstone as the core testing platform because it is a commercial, cloud-hosted solution requiring a paid subscription and external infrastructure dependency. While it offers comprehensive FHIR conformance testing, it doesn't fit a lightweight, self-hosted capstone project, and relying on an external cloud platform would also introduce data-handling concerns around uploading test data to a third-party service. HAPI FHIR's open-source, embeddable nature better suited a solution that needs to run directly within MedTalk's own development and deployment pipeline.

W7-3 · Revise your figure — version 1 to version 2

Logged late · 2 min · 4/4 required parts · 13 Sept, 19:46

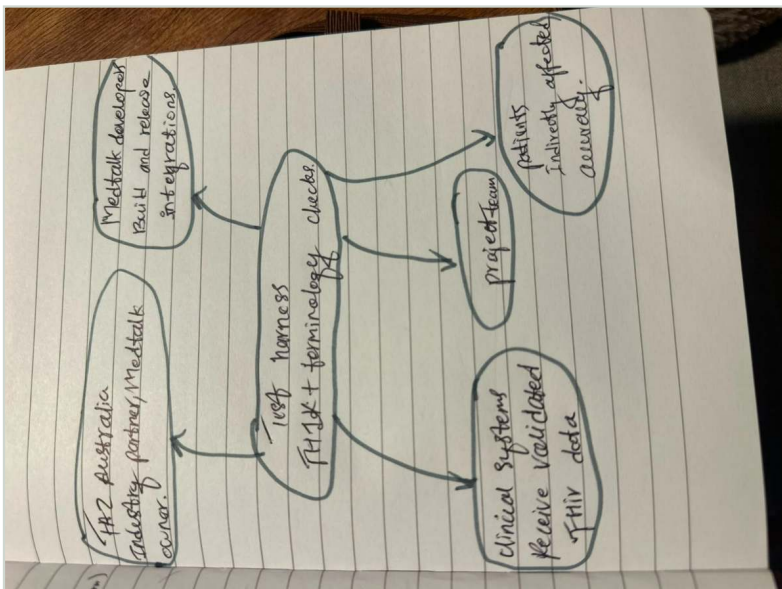
What you changed and why

In version 1, my stakeholder map only labelled each party's role but didn't show the direction of influence or dependency between them. On review, I realised a marker or teammate looking at the map couldn't tell who depends on whom — for example, that clinical systems only receive output after the harness validates it, rather than being an active input source. In version 2, I added directional arrows to show the actual flow of information (requirements in from FAZ Australia and developers, validated data out to clinical systems and patients), and repositioned the project team beneath the harness to reflect their operational role in running it, rather than treating them as just another peripheral stakeholder.

Whose feedback changed it

My own review, after re-reading the figure against the written stakeholder section and noticing the missing direction of flow.

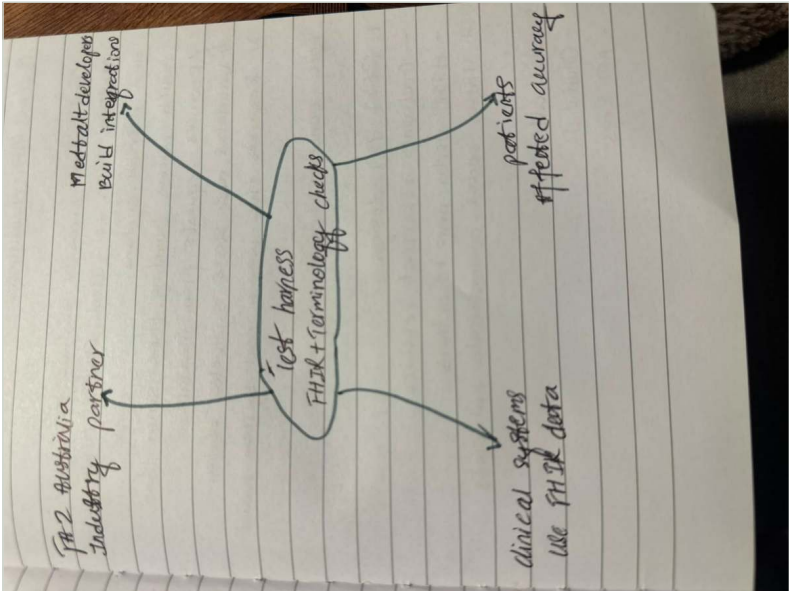
W7-3-01 · 13 SEPT 2026



Section: section 1 · Task: Revise your figure – version 1 to version 2

What it demonstrates: revised diagram

W7-3-02 · 13 SEPT 2026



Section: section 1 · Task: Revise your figure – version 1 to version 2

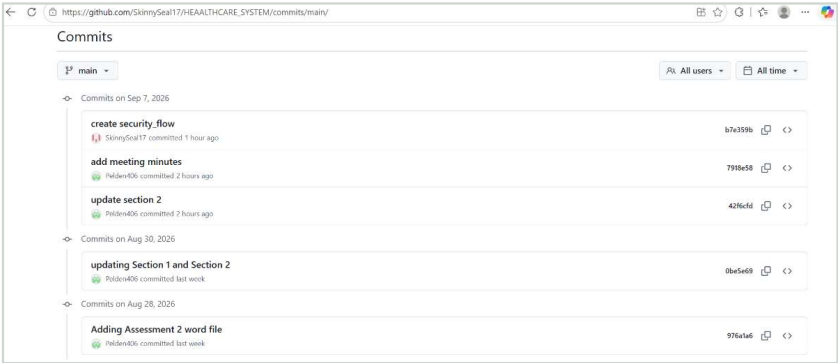
What it demonstrates: v1

W7-4 · Log a real GitHub action

Logged late · 4 min · 4/4 required parts · 07 Sept, 11:05

DATE	ISSUE / BRANCH / COMMIT REFERENCE	WHAT YOU DID	LINK	OUTCOME
30 Aug	Commit 0be5e69	Updated Section 1 and Section 2 of the Assessment 2 report (Pelden406)	github.com/SkinnySeal17/HEALTHCARE_SYSTEM/commit/0be5e69	Committed successfully, report updated

W7-4-01 · 07 SEPT 2026



Section: — · Task: Log a real GitHub action

What it demonstrates: github commit

W7-5 · Record one technical decision

Logged late · 6 min · 4/4 required parts · 07 Sept, 11:10

Decision

Use HAPI FHIR (open-source, Java-based) as the core validation framework for the test harness, rather than a commercial platform (AEGIS Touchstone)

Context — what forced the decision

The team needed to select a technical foundation for the FHIR conformance validation component before implementation

could begin (HITH-35/36). AEGIS Touchstone offered comprehensive conformance testing but required a paid subscription and cloud hosting, which conflicted with the project's need for a lightweight, self-hosted solution that could run locally in a capstone timeframe. Building a fully custom validator from scratch was considered but would consume significant development time better spent on the terminology validation layer, which is the project's more novel contribution.

Options considered

AEGIS Touchstone (commercial, cloud-based) — comprehensive but adds licensing cost, cloud dependency, and less control over customisation for our specific terminology-checking needs.

Fully custom-built validator (build from scratch) — maximum control, but high development risk and time cost within a single-semester capstone, duplicating functionality that already exists in mature open-source tools.

HAPI FHIR (open-source, Apache 2.0) — free, embeddable directly into our own codebase, actively maintained, and used widely enough that documentation and community support are strong.

Consequence, including what gets harder

This locks the team into Java-based tooling and HAPI's specific validation behaviour, meaning any future terminology or profile customisation must work within HAPI's extension model. It also means the team is responsible for hosting and maintaining the validator ourselves rather than relying on a vendor-supported platform, which could make long-term maintenance harder after the capstone ends.

W7-6 · Weekly reflection

Logged late · 5 min · 4/4 required parts · 07 Sept, 11:18

What did I personally complete this week?

This week I made and documented the technical decision to use HAPI FHIR as the core validation framework (weighing it against AEGIS Touchstone and a custom build), built a working proof-of-concept script that validates FHIR Patient resource conformance and SNOMED CT terminology codes, and logged a real GitHub commit updating Sections 1 and 2 of the Assessment 2 report. I also mapped the component to ISO/IEC 27001:2022 Annex A controls (A.8.29, A.8.28, A.5.15).

What did I learn about the problem or the technology?

I learned that FHIR's structural validation alone doesn't catch every real-world error — for example, it accepted an invalid gender code but correctly rejected a malformed date, showing that conformance checking and terminology validation genuinely need to be separate, deliberate layers rather than one combined check. This reinforced why our two-layer harness design (conformance + terminology) is necessary rather than redundant.

What decision did I influence, and why?

I influenced the choice of HAPI FHIR over AEGIS Touchstone and a fully custom-built validator, prioritising development speed and cost over the polish of a commercial platform. This kept the team within a feasible scope for a single-semester capstone while still giving us enough control to add our own terminology-checking logic on top.

What will I finish before the next class?

I'll extend the proof-of-concept to run against a small batch of sample FHIR resources rather than single hardcoded examples, and start drafting the security control mapping into the full Assessment 2 report so it's ready for team review alongside Charity's and Rikesh's Week 7 sections.

Week 8

6 of 7 entries logged · 34 minutes recorded · 8 evidence items · 90% complete

W8-1 · Show your component running

Logged late · 5 min · 4/4 required parts · 13 Sept, 19:31

What you built or configured

I built a working proof-of-concept component of the test harness in Python, using the `fhir.resources` library (FHIR R4-compliant, built on the same validation model as HAPI FHIR) to check structural conformance of Patient resources, and a config file (`snomed_reference_set.json`) representing a local SNOMED CT reference set for terminology validation. The script runs four tests: a valid Patient resource (passes conformance), an invalid resource with a malformed birthDate (correctly fails with a specific error), a valid SNOMED CT code lookup (passes), and an invalid/mismapped code (correctly fails). This demonstrates the harness's two core validation layers FHIR conformance and terminology accuracy working together exactly as scoped in the proposed architecture.

How to run or reproduce it

Install the dependency with `pip install fhir.resources`. Place `fhir_terminology_harness.py` and `snomed_reference_set.json` in the same folder, then run `python3 fhir_terminology_harness.py` from the terminal. The console output shows four labelled test results (PASS/FAIL) for FHIR structural conformance and SNOMED CT terminology validation.

W8-1-01 · 13 SEPT 2026

```
PS C:\Users\pelde\OneDrive\Desktop> python fhir_terminology_harness.py
=== TEST 1: FHIR Conformance Check (Valid Patient) ===
PASS - Resource is FHIR R4 conformant

=== TEST 2: FHIR Conformance Check (Invalid Patient - malformed birthDate) ===
FAIL - Value error, Date value string does not match spec regex. (field: ('birthDate',))

=== TEST 3: Terminology Validation (Valid SNOMED CT code) ===
PASS - 44054006 is a valid SNOMED-CT code (Type 2 diabetes mellitus)

=== TEST 4: Terminology Validation (Invalid/mismapped code) ===
FAIL - 99999999 not found in approved SNOMED-CT value set
PS C:\Users\pelde\OneDrive\Desktop>
```

Section: Section 4 Architecture · Task: Show your component running

What it demonstrates: output

W8-1-02 · 13 SEPT 2026

```
PS C:\Users\pelde\OneDrive\Desktop> pip install fhir.resources
Collecting fhir.resources
  Downloading fhir_resources-8.3.0-py2.py3-none-any.whl.metadata (49 kB)
Collecting fhir-core>=1.1.9 (from fhir.resources)
  Downloading fhir_core-1.1.10-py2.py3-none-any.whl.metadata (12 kB)
Collecting pydantic<3.0,>=2.7.4 (from fhir-core==1.1.9->fhir.resources)
  Downloading pydantic-2.13.5-py3-none-any.whl.metadata (118 kB)
Collecting annotated-types<0.8.0-py3-none-any.whl.metadata (15 kB)
  Downloading annotated_types-0.8.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.46.5 (from pydantic<3.0,>=2.7.4->fhir-core==1.1.9->fhir.resources)
  Downloading pydantic_core-2.46.5-cp314-cp314-win_and64.whl.metadata (6.7 kB)
Collecting typing-extensions>=4.14.1 (from pydantic<3.0,>=2.7.4->fhir-core==1.1.9->fhir.resources)
  Downloading typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)
Collecting typing-inspection>=0.4.2 (from pydantic<3.0,>=2.7.4->fhir-core==1.1.9->fhir.resources)
  Downloading typing_inspection-0.4.4-py3-none-any.whl.metadata (2.6 kB)
Downloading fhir_resources-8.3.0-py2.py3-none-any.whl (2.3 MB)
  2.3/2.3  5.6 MB/s  0:00:00
Downloading fhir_core-1.1.10-py2.py3-none-any.whl (30 kB)
Downloading pydantic-2.13.5-py3-none-any.whl (472 kB)
  2.0/2.0  12.8 MB/s  0:00:00
Downloading pydantic_core-2.46.5-cp314-cp314-win_and64.whl (2.0 MB)
  2.0/2.0  12.8 MB/s  0:00:00
Downloading annotated_types-0.8.0-py3-none-any.whl (13 kB)
Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
Downloading typing_inspection-0.4.4-py3-none-any.whl (14 kB)
Installing collected packages: typing-extensions, annotated-types, typing-inspection, pydantic-core, pydantic, fhir-core, fhir.resources
Successfully installed annotated-types-0.8.0 fhir-core-1.1.10 fhir.resources-8.3.0 pydantic-2.13.5 pydantic-core-2.46.5 typing-extensions-4.16.0 typing-inspection-0.4.4
```

Section: Section 4 Architecture · Task: Show your component running

What it demonstrates: output

W8-1B · Upload your sitemap, wireframe and high-fidelity design

Logged late · 9 min · 7/7 required parts · 13 Sept, 19:38

Tool used

Sitemap — the screens or pages in your solution and how they connect

The sitemap shows the core navigation of the test harness: users log in and land on a Dashboard, which links to four main screens — Run Validation (where FHIR files and reference sets are uploaded), Results (pass/fail outcomes and error logs), Reports (exportable validation history), and Reference Set Config (manage SNOMED CT/ICD-10 mappings). Results links further into a Detailed Result View for drilling into a specific validation failure. This structure keeps the workflow linear: configure, run, review, and export matching how a developer would use the tool before deploying a MedTalk integration.

Wireframe — which screen you wireframed and what a user can do on it

I wireframed the Run Validation screen, the harness's core interaction point. On it, a user can upload a FHIR resource file, select a clinical terminology reference set from a dropdown, toggle which checks to run (FHIR conformance and/or terminology mapping), trigger the validation with a button, and view a placeholder list of pass/fail results below. The layout is deliberately low-fidelity boxes and labels only to confirm the screen's structure and flow before adding visual design, ensuring the key actions (upload, configure, run, review) are all present and logically ordered.

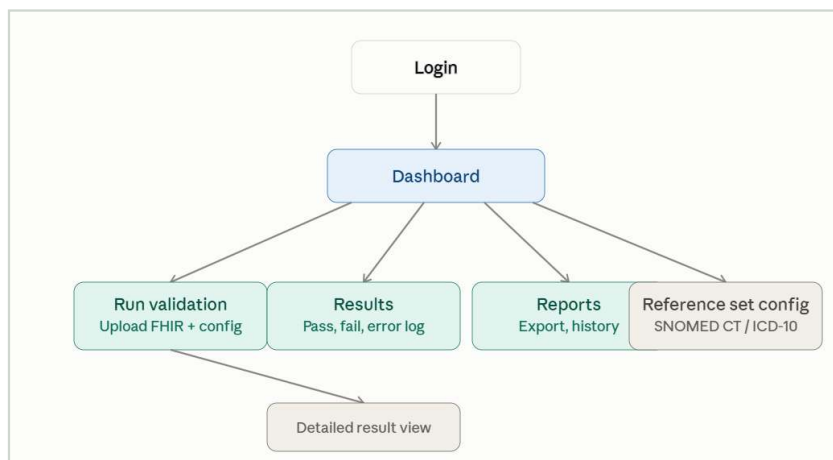
High-fidelity — what changed from the wireframe, and why

The high-fidelity version keeps the exact same layout and elements as the wireframe upload area, reference set dropdown, two checkboxes, run button, and results list but adds real visual design: a header with the app name and user avatar, an upload icon with drag-and-drop styling, a solid accent-coloured "Run validation" button, and colour-coded pass/fail icons (green check, red cross) with real result text. These changes make the interface feel functional and readable rather than just structural, while preserving the wireframe's information architecture exactly, since a mismatch between the two is what loses marks here.

Which parts of these three artefacts did you personally produce?

I personally designed and produced all three artefacts the sitemap, the wireframe, and the high-fidelity screen for the Run Validation flow, since as Project Lead and Business Analyst-Documentation Lead the harness's core UX design as part of the Proposed Solution Architecture (HITH-35).

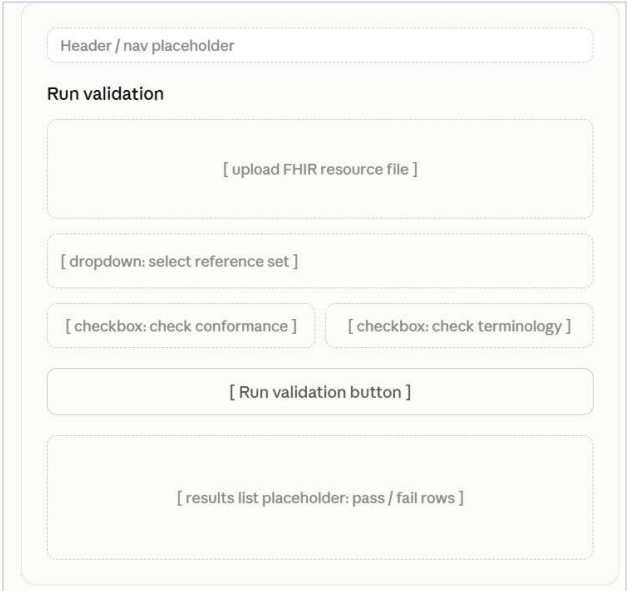
W8-1B-01 · 13 SEPT 2026



Section: Section 4 Architecture · Task: Upload your sitemap, wireframe and high-fidelity design

What it demonstrates: sitemap

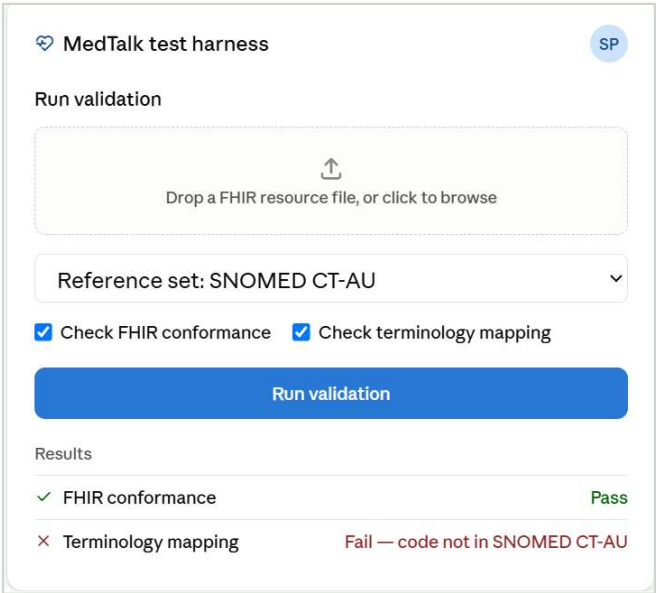
W8-1B-02 · 13 SEPT 2026



Section: Section 4 Architecture · Task: Upload your sitemap, wireframe and high-fidelity design

What it demonstrates: low-fidelity wireframe

W8-1B-03 · 13 SEPT 2026



Section: Section 4 Architecture · Task: Upload your sitemap, wireframe and high-fidelity design

What it demonstrates: high-fidelity wireframe

W8-2 · Test it and record the results

Logged late · 6 min · 4/4 required parts · 13 Sept, 19:41

TEST ID	WHAT IS BEING TESTED	INPUT / STEPS	EXPECTED	ACTUAL	PASS / FAIL
T-01	FHIR structural conformance - valid resource	Run harness against a valid Patient resource (correct name, gender, birthDate format)	Resource passes conformance check	"PASS - Resource is FHIR R4 conformant"	Pass
T-02	FHIR structural conformance - invalid date format	Run harness against a Patient resource with birthDate "12-05-1998" (wrong format, should be YYYY-MM-DD)	Resource fails conformance check with a specific field error	"FAIL - Value error, Date value string does not match spec regex. (field: ('birthDate'.))"	Fail

TEST ID	WHAT IS BEING TESTED	INPUT / STEPS	EXPECTED	ACTUAL	PASS / FAIL
T-03	Terminology validation - valid SNOMED CT code	Check code 44054006 against local SNOMED-CT reference set	Code is recognised as valid	"PASS - 44054006 is a valid SNOMED-CT code (Type 2 diabetes mellitus)"	Pass

W8-2-01 · 13 SEPT 2026

```
PS C:\Users\pelde\OneDrive\Desktop> python fhir_terminology_harness.py
=== TEST 1: FHIR Conformance Check (Valid Patient) ===
PASS - Resource is FHIR R4 conformant

=== TEST 2: FHIR Conformance Check (Invalid Patient - malformed birthDate) ===
FAIL - Value error, Date value string does not match spec regex. (field: ('birthDate',))

=== TEST 3: Terminology Validation (Valid SNOMED CT code) ===
PASS - 44054006 is a valid SNOMED-CT code (Type 2 diabetes mellitus)

=== TEST 4: Terminology Validation (Invalid/mismapped code) ===
FAIL - 99999999 not found in approved SNOMED-CT value set
PS C:\Users\pelde\OneDrive\Desktop>
```

Section: Section 4 Architecture · Task: Test it and record the results

What it demonstrates: output

W8-2-02 · 13 SEPT 2026

```
PS C:\Users\pelde\OneDrive\Desktop> pip install fhir.resources
Collecting fhir.resources
  Downloading fhir_resources-8.3.0-py2.py3-none-any.whl.metadata (49 kB)
Collecting fhir-core==1.1.9 (from fhir-resources)
  Downloading fhir_core-1.1.10-py2.py3-none-any.whl.metadata (12 kB)
Collecting pydantic<3.0,>=2.7.4 (from fhir-core==1.1.9->fhir-resources)
  Downloading pydantic-2.13.5-py3-none-any.whl.metadata (110 kB)
Collecting annotated-types>=0.6.0 (from pydantic<3.0,>=2.7.4->fhir-core==1.1.9->fhir-resources)
  Downloading annotated_types-0.8.0-py3-none-any.whl.metadata (15 kB)
Collecting pydantic-core==2.46.5 (from pydantic<3.0,>=2.7.4->fhir-core==1.1.9->fhir-resources)
  Downloading pydantic_core-2.46.5-cp314-cp314-win_and64.whl.metadata (6.7 kB)
Collecting typing-extensions>=4.14.1 (from pydantic<3.0,>=2.7.4->fhir-core==1.1.9->fhir-resources)
  Downloading typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)
Collecting typing-inspection>=0.4.2 (from pydantic<3.0,>=2.7.4->fhir-core==1.1.9->fhir-resources)
  Downloading typing_inspection-0.4.4-py3-none-any.whl.metadata (2.6 kB)
Downloading fhir_resources-8.3.0-py2.py3-none-any.whl (2.3 MB)
  2.3/2.3  5.6 MB/s  0:00:00
Downloading fhir_core-1.1.10-py2.py3-none-any.whl (30 kB)
Downloading pydantic-2.13.5-py3-none-any.whl (472 kB)
Downloading pydantic_core-2.46.5-cp314-cp314-win_and64.whl (2.0 MB)
  2.0/2.0  12.8 MB/s
Downloading annotated_types-0.8.0-py3-none-any.whl (13 kB)
Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
Downloading typing_inspection-0.4.4-py3-none-any.whl (14 kB)
Installing collected packages: typing-extensions, annotated-types, typing-inspection, pydantic-core, pydantic, fhir-core, fhir.resources
Successfully installed annotated-types-0.8.0 fhir-core-1.1.10 fhir.resources-8.3.0 pydantic-2.13.5 pydantic-core-2.46.5 typing-extensions-4.16.0 typing-inspection-0.4.4
```

Section: – · Task: Test it and record the results

What it demonstrates: output

W8-3 · Map your work to a security framework

Logged late · 1 min · 3/3 required parts · 13 Sept, 19:41

Framework used

ISO/IEC 27001:2022 Annex A

CONTROL REFERENCE	CONTROL NAME	HOW YOUR COMPONENT ADDRESS	FULL / PARTIAL / PLANNED
A.8.29	Security testing in development and acceptance	The harness runs automated conformance and terminology checks against sample FHIR resources before they would be accepted into MedTalk's release pipeline, catching structural and mapping errors pre-deployment	Partial — checks work standalone; not yet wired into an actual CI/acceptance pipeline
A.8.28	Secure coding	The component validates that incoming FHIR resources conform to the R4 specification and rejects malformed data (e.g. invalid date formats), reducing the risk of malformed or unsafe data being processed downstream	Partial — covers structural/data validation only, not full secure coding practices (e.g. no static analysis, no dependency scanning yet)

CONTROL REFERENCE	CONTROL NAME	HOW YOUR COMPONENT ADDRESS	FULL / PARTIAL / PLANNED
A.5.15	Access control	The current PoC runs as a standalone script with no login, roles, or access restrictions – anyone with file access can run it	Planned – access control was scoped for a later development phase, not built into this PoC

W8-4 · Ethics, privacy, legal and AI governance

Logged late · 1 min · 3/3 required parts · 13 Sept, 19:41

Personal data your solution touches, and the obligation that applies

The test harness handles FHIR resources containing patient identifiers (name, birthDate) and clinical terminology codes (e.g. SNOMED CT diagnosis codes) drawn from MedTalk's AI-generated clinical documentation. This data falls under the Australian Privacy Principles (APPs) under the Privacy Act 1988, specifically APP 6 (use and disclosure) and APP 11 (data security), since diagnosis-linked patient data is health information and classified as sensitive information requiring a higher standard of protection. Because of this, the harness must not use real, identifiable patient records for testing; it uses synthetic or de-identified test data instead, consistent with the requirement to minimise handling of sensitive personal information wherever the same testing outcome can be achieved without it.

AI use in this project and how it was governed

I used Claude generate a Python proof-of-concept for the validation logic, and search for and summarise literature on FHIR/terminology validation tools. All AI-assisted outputs were reviewed and edited by me before inclusion, and technical claims (tool features, licensing, paper findings) were checked against primary sources rather than accepted at face value. AI was not used to generate or fabricate test data, citations, or results — the validation script's test cases and outputs were run and verified locally. This aligns with CIHE's expectation that AI is used as a drafting and research aid, with the student remaining accountable for accuracy and originality of the final submission.

One ethical or legal risk you cannot fully mitigate

The harness's synthetic test data may not fully replicate the messiness or edge cases of real de-identified patient data, meaning a validation pass in testing doesn't guarantee the same reliability once the harness is used against MedTalk's actual production-scale, real-world clinical data.

W8-5 · Own a slide

Logged late · 9 min · 5/5 required parts · 13 Sept, 19:41

Slide title and number you are responsible for

Slide 3 & 4 — "Business Context & Problem Statement" and "Goals, Scope & Stakeholders" (Section 1: Introduction).

The three points you will make

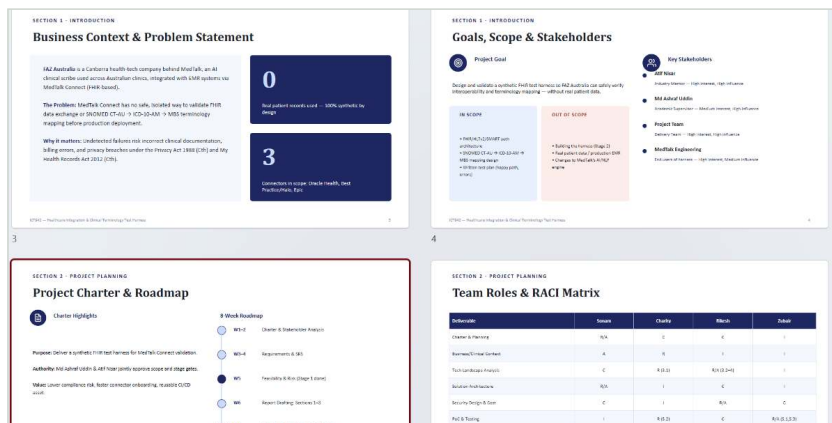
1. FAZ Australia's MedTalk Connect currently has no safe, isolated way to test FHIR interoperability and terminology mapping before deployment, forcing reliance on manual checks against live-adjacent systems.
2. this creates real risk: incorrect SNOMED CT-AU to ICD-10 mappings or non-conformant FHIR messages could compromise patient data integrity and breach Australian healthcare compliance standards.
3. our project's goal is to design and validate a synthetic FHIR test harness that lets FAZ Australia safely verify interoperability and terminology mapping without ever touching real patient data, with a scope covering FHIR/HL7v2/SMART auth architecture and SNOMED CT-AU to ICD-10 mapping validation.

A question you expect from the audience, and your answer

Question: "Why use synthetic data instead of real, de-identified patient records for testing?"

Answer: Synthetic data avoids any risk of exposure or compliance breach entirely, since there's no real patient information involved at any stage. It also means the harness can be run and shared freely during development without requiring clinical governance approval, which would be a significant time and administrative cost for a single-semester capstone project.

W8-5-01 · 07 SEPT 2026



Section: section 1 • Task: Own a slide

What it demonstrates: slide 3,4,5,6

W8-6 • Limitations, peer cross-check and final reflection

Draft only • 3 min • 2/7 required parts

Honest limitations of your contribution

My contributions this week (Business Context/Problem Statement, Project Charter/RACI, and Risk Register input) are strong on planning and documentation but haven't yet been tested against real feedback from FAZ Australia or validated by the technical build. The problem statement and scope decisions are based on my own analysis and the literature I reviewed, not direct confirmation from the industry partner, so some assumptions (e.g. the extent of manual validation effort at MedTalk) may need revising once we get more detailed input. The RACI matrix also reflects planned responsibilities rather than tested workload balance, so it may need adjusting once the team is further into implementation.

What did I personally complete this week?

I completed drafts of my assigned tasks: the Business Context, Problem Statement, Goals/Objectives, Scope, and Stakeholders section, and the Project Charter with Roadmap/Gantt Chart and RACI Matrix. Both are marked In Review on Jira, ready for team feedback before the next class.

What did I learn about the problem or the technology?

Manual validation of HL7 FHIR integrations and SNOMED CT mappings is a well-documented weak point in healthcare systems generally, not just for MedTalk. Literature showed measurable manual mapping error rates and confirmed that automated FHIR validation tools meaningfully improve compliance, reinforcing that our proposed test harness addresses a real, recognised gap.

What decision did I influence, and why?

As Project Lead, I influenced scoping the project around FHIR conformance and terminology validation only, excluding real-time production monitoring, due to time constraints and the ethical risk of using real patient data without clinical governance approval.

What will I finish before the next class?

I'll finalise the Business Context/Problem Statement section and the Project Charter/RACI Matrix based on any team feedback, and begin supporting Rikesh and Charity's Section 3/4 drafts where they overlap with scope and stakeholder requirements.

Literature and evidence tracker

WK	SOURCE	SEARCH METHOD / ORIGIN	KEY INFORMATION COLLECTED	HOW IT WAS USED
6	JMIR Walonoski et al. (2018), Validation and Testing of Fast Healthcare Interoperability Resources Standards Compliance: Data Analysis	Google Scholar	Found that regular use of FHIR validation/testing tools (Crucible, Touchstone) is strongly correlated with improved server compliance with the HL7 FHIR spec	Justifies the need for an automated test harness — supports Section 3 (Technology and Market Survey) and the project rationale
6	Ohlsen, Hofer & Ingenerf (2025), A Validation Tool (VaPCE) for Postcoordinated SNOMED CT Expressions	Google Scholar	Describes a tool using FHIR's \$validate-code operation with a terminology server to detect and explain SNOMED CT syntactic/semantic errors	Direct technical precedent for the test harness's terminology validation module (Section 4 – Solution Design)
6	Vikström et al. (2007), Mapping the categories of the Swedish primary health care version of ICD-10 to SNOMED CT concepts	Google Scholar	Found intercoder reliability of only 83% when manually mapping ICD-10 to SNOMED CT, showing manual mapping is error-prone	Evidence for the problem statement — quantifies risk of manual terminology mapping
7	HAPI FHIR Validator (Smile CDR / HL7 community — open source, Apache 2.0 Licence)	HL7 FHIR Confluence, "Using the FHIR Validator," https://confluence.hl7.org/display/FHIR/Using+the+FHIR+Validator	Validates FHIR resources against StructureDefinitions and profiles; uses terminology services to check codes (e.g. SNOMED CT, LOINC) against ValueSets; runs locally or via CLI, so it can be embedded directly into a CI/testing pipeline	In-memory terminology checking cannot validate external CodeSystems (like full SNOMED CT or LOINC) without connecting to an external terminology server, adding setup complexity
7	AEGIS Touchstone (AEGIS.net, Inc. — commercial, cloud-based subscription platform (free tier available))	AEGIS.net, "Touchstone – Anchor of the AEGIS testing ecosystem," https://touchstone.com/	Automated, cloud-based conformance and interoperability testing against the FHIR specification; supports custom TestScripts and continuous conformance measurement, useful as a benchmark for our test harness design	Commercial licensing cost and cloud dependency make it unsuitable as a directly embedded solution for a capstone-scale, self-hosted test harness
7	Ontoserver (CSIRO) (CSIRO (Australian e-Health Research Centre) — commercial licence, with free academic-use tier)	Ontoserver, "The gold-standard of FHIR terminology servers," https://www.ontoserver.csiro.au/site/our-solutions/ontoserver/	Purpose-built FHIR terminology server with deep SNOMED CT-AU support; provides \$validate-code, \$expand, and \$subsumes operations for verifying terminology mappings; underpins Australia's National Clinical Terminology Service	Requires a formal licensing agreement and infrastructure to host, which adds cost and setup overhead not ideal for a lightweight capstone proof-of-concept
8	ISO/IEC 27001:2022 Annex A A.8.29	framework document	Security testing in development and acceptance	The harness runs automated conformance and terminology checks against sample FHIR resources before they would be accepted into MedTalk's release pipeline, catching structural and mapping errors pre-deployment
8	ISO/IEC 27001:2022 Annex A A.8.28	framework document	Secure coding	The component validates that incoming FHIR resources conform to the R4 specification and rejects malformed data (e.g. invalid date formats), reducing the risk of malformed or unsafe data being processed downstream

WK	SOURCE	SEARCH METHOD / ORIGIN	KEY INFORMATION COLLECTED	HOW IT WAS USED
8	ISO/IEC 27001:2022 Annex A A.5.15	framework document	Access control	The current PoC runs as a standalone script with no login, roles, or access restrictions – anyone with file access can run it

GitHub and project-management activity

DATE	PLATFORM	REF	ACTION	LINK	OUTCOME
30 Aug	GitHub	Commit 0be5e69	Updated Section 1 and Section 2 of the Assessment 2 report (Pelden406)	github.com/SkinnySeal17/HEALTHCARE_SYSTEM/commit/0be5e69	Committed successfully, report updated
Wk6	Board	Task created:	Created or took ownership		
Wk6	Board	HITH-31 – Business Context, Problem Stat	Created or took ownership		
Wk6	Board	HITH-32 – Project Charter, Roadmap/Gantt	Created or took ownership		
Wk6	Board	HITH-33 – Current State Assessment: As-I	Created or took ownership		
Wk6	Board	HITH-34 – Technology and Market Survey	Created or took ownership		
Wk6	Board	HITH-35 – Proposed Solution Architecture	Created or took ownership		
Wk6	Board	HITH-36 – Technology Stack and Justifica	Created or took ownership		
Wk6	Board	HITH-37 – Sprint Plan	Created or took ownership		
Wk6	Board	Task Assigned(ownership):	Created or took ownership		
Wk6	Board	HITH-31 – Business Context, Problem Stat	Created or took ownership		
Wk6	Board	HITH-32 – Project Charter, Roadmap/Gantt	Created or took ownership		

Evidence index

NO .	COLLECTED	SECTION	TASK	WHAT IT DEMONSTRATES
W6-2-01	24 Aug 2026	Section 3 Technology and Market Survey) & Introduction/rationale	Literature search log	the screenshot shows the file that i have searched
W6-2-02	31 Aug 2026	Section 4.1 Porposed souldion Architecture	Literature search log	this image shows the file related to the solution design
W6-2-03	31 Aug 2026	section 1.1 Business Context and Problem Statement	Literature search log	This image support the business context
W6-3-01	31 Aug 2026	All the section	Claim your tasks on the board	jira tasks backlog
W6-4-01	13 Sept 2026	—	Draw the first figure	diagram

NO .	COLLECTED	SECTION	TASK	WHAT IT DEMONSTRATES
W7-1-01	31 Aug 2026	section 3	Compare three existing tools	Shows the validator is open source (Apache 2.0), distributed as a standalone Java tool, and documents its know
W7-1-02	31 Aug 2026	section 3	Compare three existing tools	Demonstrates it is a cloud-based commercial FHIR conformance testing platform with automated TestScript execut
W7-1-03	31 Aug 2026	section 3	Compare three existing tools	Demonstrates it is a commercial, SNOMED CT-focused FHIR terminology server with validation and search features
W7-3-01	13 Sept 2026	section 1	Revise your figure — version 1 to version 2	revised diagram
W7-3-02	13 Sept 2026	section 1	Revise your figure — version 1 to version 2	v1
W7-4-01	07 Sept 2026	—	Log a real GitHub action	github commit
W8-1-01	13 Sept 2026	Section 4 Architecture	Show your component running	output
W8-1-02	13 Sept 2026	Section 4 Architecture	Show your component running	output
W8-1B-01	13 Sept 2026	Section 4 Architecture	Upload your sitemap, wireframe and high-fidelity design	sitemap
W8-1B-02	13 Sept 2026	Section 4 Architecture	Upload your sitemap, wireframe and high-fidelity design	low-fidelity wireframe
W8-1B-03	13 Sept 2026	Section 4 Architecture	Upload your sitemap, wireframe and high-fidelity design	high-fidelity wireframe
W8-2-01	13 Sept 2026	Section 4 Architecture	Test it and record the results	output
W8-2-02	13 Sept 2026	—	Test it and record the results	output
W8-5-01	07 Sept 2026	section 1	Own a slide	slide 3,4,5,6

VERIFICATION

Time log

Each entry below was recorded automatically while the task was open in the portfolio tool. The timer pauses after three minutes without activity.

TASK	SESSION START	DURATION	CONTEXT
S-2	24 Aug, 10:28	00:24	In class
S-2	24 Aug, 10:30	00:30	In class
W6-1	24 Aug, 10:33	00:30	Outside class
W6-1	24 Aug, 10:34	00:06	Outside class
W6-2	24 Aug, 10:34	00:06	Outside class
W6-4	24 Aug, 10:34	00:26	Outside class
W6-5	24 Aug, 10:34	00:06	Outside class
W6-1	24 Aug, 10:35	00:08	Outside class
W6-2	24 Aug, 10:43	00:30	Outside class
W6-2	24 Aug, 10:43	00:31	Outside class
W6-2	24 Aug, 10:44	00:30	Outside class
W6-2	24 Aug, 10:44	00:31	Outside class
W6-2	24 Aug, 10:45	00:31	Outside class
W6-2	24 Aug, 10:45	00:31	Outside class
W6-2	24 Aug, 10:46	00:30	Outside class
W6-2	24 Aug, 10:46	00:31	Outside class
W6-2	24 Aug, 10:47	00:30	Outside class
W6-2	24 Aug, 10:47	00:31	Outside class
W6-2	24 Aug, 10:48	00:30	Outside class
W6-2	24 Aug, 10:48	00:31	Outside class
W6-2	24 Aug, 10:49	00:31	Outside class
W6-2	24 Aug, 10:49	00:31	Outside class
W6-2	24 Aug, 10:50	00:31	Outside class
W6-2	24 Aug, 10:50	00:31	Outside class
W6-2	24 Aug, 10:51	00:30	Outside class
W6-2	24 Aug, 10:51	00:31	Outside class
W6-2	24 Aug, 10:52	00:30	Outside class
W6-3	24 Aug, 11:07	00:08	Outside class
W6-5	24 Aug, 11:07	00:13	Outside class
W6-2	24 Aug, 11:08	00:09	Outside class
W6-2	24 Aug, 11:08	00:30	Outside class
W6-2	24 Aug, 11:08	00:27	Outside class
W6-2	24 Aug, 11:10	00:30	Outside class

TASK	SESSION START	DURATION	CONTEXT
W6-2	31 Aug, 08:26	00:30	In class
W6-2	31 Aug, 08:27	00:22	In class
W6-3	31 Aug, 08:27	00:30	In class
W6-3	31 Aug, 08:28	00:30	In class
W6-3	31 Aug, 08:28	00:31	In class
W6-3	31 Aug, 08:29	00:15	In class
W6-5	31 Aug, 08:41	00:30	In class
W6-5	31 Aug, 08:41	00:31	In class
W6-5	31 Aug, 08:42	00:30	In class
W6-5	31 Aug, 08:42	00:31	In class
S-2	31 Aug, 08:43	00:30	Outside class
W6-5	31 Aug, 08:43	00:09	In class
S-2	31 Aug, 08:44	00:31	Outside class
S-2	31 Aug, 08:44	00:31	Outside class
S-2	31 Aug, 08:45	00:31	Outside class
S-2	31 Aug, 08:45	00:31	Outside class
S-2	31 Aug, 08:46	00:31	Outside class
S-2	31 Aug, 08:46	00:31	Outside class
S-2	31 Aug, 08:47	00:18	Outside class
W6-2	31 Aug, 08:50	00:27	In class
W6-3	31 Aug, 08:50	00:18	In class
W6-4	31 Aug, 08:50	00:11	In class
W6-5	31 Aug, 08:51	00:23	In class
W6-2	31 Aug, 09:26	00:30	In class
W6-2	31 Aug, 09:27	00:31	In class
W6-2	31 Aug, 09:27	00:31	In class
W6-2	31 Aug, 09:28	00:31	In class
W6-2	31 Aug, 09:28	00:31	In class
W6-2	31 Aug, 09:29	00:30	In class
W6-2	31 Aug, 09:29	00:31	In class
W6-2	31 Aug, 09:30	00:30	In class
W6-2	31 Aug, 09:30	00:30	In class
W6-2	31 Aug, 09:31	00:31	In class
W6-2	31 Aug, 09:31	00:31	In class
W6-2	31 Aug, 09:32	00:30	In class
W6-2	31 Aug, 09:32	00:31	In class
W6-2	31 Aug, 09:33	00:31	In class

TASK	SESSION START	DURATION	CONTEXT
W6-3	31 Aug, 09:33	00:30	In class
W6-3	31 Aug, 09:34	00:27	In class
W6-4	31 Aug, 09:34	00:21	In class
W6-5	31 Aug, 09:35	00:31	In class
W6-5	31 Aug, 09:35	00:30	In class
W6-5	31 Aug, 09:36	00:31	In class
W6-5	31 Aug, 09:36	00:31	In class
W6-3	31 Aug, 09:37	00:27	In class
W6-5	31 Aug, 09:37	00:14	In class
W6-5	31 Aug, 09:38	00:30	In class
W6-5	31 Aug, 09:38	00:31	In class
W6-5	31 Aug, 09:39	00:31	In class
W6-5	31 Aug, 09:39	00:31	In class
W6-5	31 Aug, 09:40	00:31	In class
W6-5	31 Aug, 09:40	00:31	In class
W6-5	31 Aug, 09:41	00:30	In class
W6-5	31 Aug, 09:41	00:31	In class
W6-5	31 Aug, 09:42	00:31	In class
W6-5	31 Aug, 09:42	00:31	In class
W6-5	31 Aug, 09:43	00:31	In class
W6-5	31 Aug, 09:43	00:30	In class
W6-5	31 Aug, 09:44	00:31	In class
W6-5	31 Aug, 09:44	00:31	In class
W6-5	31 Aug, 09:45	00:30	In class
W6-5	31 Aug, 09:45	00:31	In class
W6-5	31 Aug, 09:46	00:31	In class
W6-5	31 Aug, 09:46	00:31	In class
W6-5	31 Aug, 09:47	00:31	In class
W6-5	31 Aug, 09:47	00:30	In class
W6-5	31 Aug, 09:48	00:30	In class
W6-5	31 Aug, 09:48	00:31	In class
W6-5	31 Aug, 09:49	00:30	In class
W6-5	31 Aug, 09:49	00:31	In class
W6-5	31 Aug, 09:50	00:20	In class
W7-1	31 Aug, 09:51	00:19	In class
W7-1	31 Aug, 09:51	00:30	In class
W7-1	31 Aug, 09:52	00:30	In class

TASK	SESSION START	DURATION	CONTEXT
W7-1	31 Aug, 09:52	00:31	In class
W7-1	31 Aug, 09:53	00:31	In class
W7-1	31 Aug, 09:53	00:31	In class
W7-1	31 Aug, 09:54	00:31	In class
W7-1	31 Aug, 09:54	00:31	In class
W7-1	31 Aug, 09:55	00:31	In class
W7-1	31 Aug, 09:55	00:31	In class
W7-1	31 Aug, 09:56	00:30	In class
W7-1	31 Aug, 09:56	00:31	In class
W7-1	31 Aug, 09:57	00:29	In class
W7-1	31 Aug, 09:57	00:30	In class
W7-1	31 Aug, 09:58	00:08	In class
W7-1	31 Aug, 10:11	00:30	In class

Weekly individual reflection

Week 6

What did I personally complete this week?

For Week 6, I have completed two key tasks: drafting the Business Context, Problem Statement, Goals/Objectives, Scope, and Stakeholders section, and developing the Project Charter, Roadmap/Gantt Chart, and RACI Matrix. These tasks required defining the cybersecurity problem for FAZ Australia's MedTalk platform, setting SMART objectives and success criteria, outlining project scope and stakeholder requirements, and formalising team roles and responsibilities to guide the project through its remaining phases.

What did I learn about the problem or the technology?

I learned that manual validation of HL7 FHIR integrations and SNOMED CT terminology mappings is a well-documented weak point in healthcare systems, not just an issue specific to MedTalk. Literature confirmed that manual terminology mapping has measurable error rates (e.g. intercoder agreement around 83% in one study), and that automated FHIR validation tools have been shown to meaningfully improve compliance and reduce errors compared to manual checks. This reinforced that our proposed test harness addresses a real, recognised gap rather than a hypothetical one, and pointed us toward using established validation approaches (e.g. FHIR's \$validate-code operation) rather than building terminology-checking logic from scratch.

What decision did I influence, and why?

As Project Lead, I influenced the decision to scope the project around FHIR conformance and terminology validation only, deliberately excluding real-time production monitoring of live patient data. This was based on time constraints and the ethical risk of accessing real patient records without clinical governance approval. I also influenced the RACI matrix structure, assigning clear ownership of feasibility, requirements, and architecture tasks to team members based on their strengths, which helped keep Week 6 deliverables on track despite parallel workstreams.

What will I finish before the next class?

Before the next class, I completed the drafts of both my assigned Week 6 tasks: the Business Context, Problem Statement, Goals/Objectives, Scope, and Stakeholders section. Tasks are now marked as In Review on Jira, ready for team feedback and integration into the full Assessment 2 report before the next submission checkpoint.

Week 7

What did I personally complete this week?

This week I made and documented the technical decision to use HAPI FHIR as the core validation framework (weighing it against AEGIS Touchstone and a custom build), built a working proof-of-concept script that validates FHIR Patient resource conformance and SNOMED CT terminology codes, and logged a real GitHub commit updating Sections 1 and 2 of the Assessment 2 report. I also mapped the component to ISO/IEC 27001:2022 Annex A controls (A.8.29, A.8.28, A.5.15).

What did I learn about the problem or the technology?

I learned that FHIR's structural validation alone doesn't catch every real-world error — for example, it accepted an invalid gender code but correctly rejected a malformed date, showing that conformance checking and terminology validation genuinely need to be separate, deliberate layers rather than one combined check. This reinforced why our two-layer harness design (conformance + terminology) is necessary rather than redundant.

What decision did I influence, and why?

I influenced the choice of HAPI FHIR over AEGIS Touchstone and a fully custom-built validator, prioritising development speed and cost over the polish of a commercial platform. This kept the team within a feasible scope for a single-semester capstone while still giving us enough control to add our own terminology-checking logic on top.

What will I finish before the next class?

I'll extend the proof-of-concept to run against a small batch of sample FHIR resources rather than single hardcoded examples, and start drafting the security control mapping into the full Assessment 2 report so it's ready for team review alongside Charity's and Rikesh's Week 7 sections.

Week 8

What did I personally complete this week?

I completed drafts of my assigned tasks: the Business Context, Problem Statement, Goals/Objectives, Scope, and

Stakeholders section, and the Project Charter with Roadmap/Gantt Chart and RACI Matrix. Both are marked In Review on Jira, ready for team feedback before the next class.

What did I learn about the problem or the technology?

Manual validation of HL7 FHIR integrations and SNOMED CT mappings is a well-documented weak point in healthcare systems generally, not just for MedTalk. Literature showed measurable manual mapping error rates and confirmed that automated FHIR validation tools meaningfully improve compliance, reinforcing that our proposed test harness addresses a real, recognised gap.

What decision did I influence, and why?

As Project Lead, I influenced scoping the project around FHIR conformance and terminology validation only, excluding real-time production monitoring, due to time constraints and the ethical risk of using real patient data without clinical governance approval.

What will I finish before the next class?

I'll finalise the Business Context/Problem Statement section and the Project Charter/RACI Matrix based on any team feedback, and begin supporting Rikesh and Charity's Section 3/4 drafts where they overlap with scope and stakeholder requirements.

Declaration

I confirm that the work recorded in this portfolio is my own individual contribution to the group project. Screenshots and links refer to work I personally performed. Where AI tools were used, this is disclosed in the Week 8 governance entry.

Signed: _____ Date: _____